
Bimodal masked language modeling for bulk RNA-seq and DNA methylation representation learning

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Abstract

Oncologists are increasingly relying on multiple modalities to model the complexity of diseases. Within this landscape, transcriptomic and epigenetic data have proven to be particularly instrumental and play an increasingly vital role in clinical applications. However, their integration into multimodal models remains a challenge, especially considering their high dimensionality. In this work, we present a novel bimodal model that jointly learns representations of bulk RNA-seq and DNA methylation leveraging self-supervision from masked language modeling. We implement an architecture that reduces the memory footprint usually attributed to purely transformer-based models when dealing with long sequences. We demonstrate that the obtained bimodal embeddings can be used to fine-tune cancer-type classification and survival models that achieve state-of-the-art performance compared to unimodal models. Furthermore, we introduce a robust learning framework that maintains downstream task performance despite missing modalities, enhancing the model’s applicability in real-world clinical settings.

1. Introduction

The growing availability of high-throughput technologies has revolutionized molecular research, generating extensive genomic, transcriptomic, and epigenomic data that hold immense potential for personalized medicine (Ho et al., 2021; Stark et al., 2019; Dai & Shen, 2022). Cancer diagnosis and prognosis thus increasingly rely on heterogeneous patient data, and the integration of these diverse data sources

remains a significant challenge, even more so when some modalities may be missing in real clinical applications. The high dimensionality of each modality often makes classic machine learning and deep learning methods ineffective for diagnostic purposes. As a result, there is a growing trend to first learn representations of the data, particularly using self-supervised approaches. In this context, foundation models have steadily emerged as powerful tools to learn effective and generalizable embeddings that can be applied to biological and clinical tasks. Trained with an unsupervised language modeling objective, they have already been applied to a wide range of omics data, including genomics (Dalla-Torre et al., 2025; Brixi et al., 2025), single-cell transcriptomics (Cui et al., 2024) or bulk RNA-seq (G  lard et al., 2025). These models extensively leverage the transformer architecture (Vaswani et al., 2017) which limits the maximum input length of the model due to the quadratic memory scaling of the attention mechanism. Recent models have developed new architectures to cope with these long-range sequences, either by integrating convolutional blocks (Avsec et al., 2021; Linder et al., 2025; Joshi et al., 2025) or state-space models (Popov et al., 2025). Among multiple studies, the Cancer Genome Atlas (TCGA, <https://portal.gdc.cancer.gov/>) is a publicly available dataset that gathers multi-omics data, in particular bulk RNA-seq and DNA methylation, which are the focus of this work. Including clinical information such as survival time and divided into 33 cohorts (or cancer types), this dataset is a popular benchmark for evaluating survival analysis and cancer-type classification methods. Survival analysis, or time-to-event prediction, aims to predict the time from diagnosis to the patient’s death from the disease using censored data. Though classically tackled with Cox regression (Cox, 1972), the Cox partial likelihood has more recently been reformulated as a loss used to train deep learning architectures (Ching et al., 2018; Katzman et al., 2018).

In this paper, we introduce *MOJO*, standing for **M**ulti-**O**mic **J**oint representation learning, which we tailor in this study for learning joint embeddings of bulk RNA-seq and DNA methylation through bimodal masked language modeling from the TCGA dataset. We leverage a multi-

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modal architecture that employs a mix of convolutional and transformer layers. We show that the embeddings learned by *MOJO* lead to state-of-the-art performance in various tasks from pan-cancer cancer-type classification, survival analysis, to cancer subtype clustering. We further present a framework that allows for a downstream task model to preserve its performance in the absence of a modality by introducing an auxiliary loss based on mutual information. Additionally, we demonstrate that *MOJO*'s gene-level aggregation makes it inherently array-agnostic, enabling robust zero-shot transfer to legacy Illumina 27k methylation platforms without any retraining. We also validate out-of-distribution generalization on two fully held-out external cohorts, confirming that its bimodal advantage persists on unseen populations. The code and model weights are publicly available at <https://github.com/instadeepai/multiomics-open-research>, with pretrained checkpoints on *Hugging Face*.

2. Conflict of interest disclosure

The author *M.G.* is employed by *InstaDeep*, which supported the development of *BulkRNABert*, which was among the models evaluated in this paper.

3. Related works

Omics representation learning Omics representations have traditionally been derived from statistical methods such as Principal Component Analysis (Jolliffe, 2002) or Non-negative Matrix Factorization (NMF) (Lee & Seung, 1999), the latter being particularly suited for RNA-seq and DNA methylation due to their positivity. Deep learning architectures such as Masked Auto-Encoders (Gross et al., 2024) or Mixture-of-Experts (Meng et al., 2023) have since been applied to learn omics representations used either for survival analysis or cancer-type predictions. In line with foundation models for single-cell transcriptomics (Cui et al., 2024; Yang et al., 2022), G  lard et al. (2025) developed a transformer-based model for bulk RNA-seq representation learning. Multi-modal integration is often performed using late integration; *i.e.*, each source is encoded separately before being aggregated, often using Kronecker product (Chen et al., 2020), element-wise operations (Vale-Silva & Rohr, 2021) or cross-attention (Garau-Luis et al., 2024). On the other hand, Benkirane et al. (2023) developed a custom integration of multi-omics data using variational auto-encoders (Kingma & Welling, 2013).

Missing modalities Improving the robustness of multi-modal models under modality absence is crucial given the sensitivity of recent deep learning architectures to missing modalities (Ma et al., 2022). Part of the literature focuses on techniques that operate at the data level, namely relying

on modality imputation (Chen et al., 2024; Zhang et al., 2021a; Ma et al., 2021). Zhi et al. (2024) propose a retrieval-augmented in-context learning framework to address the missing modalities issue in a low-data regime. Another path towards handling missing modalities lies in adjusting the model itself with, for example, model fusion (Wagner et al., 2011) or knowledge distillation (Saha et al., 2024). Training methods are also adapted to make multimodal models robust to missing modalities by employing modality dropout (Krishna et al., 2024; Nezakati et al., 2024) during training to simulate scenarios where a subset of modalities might be missing. In our work, we adapt a technique from Ramazanov et al. (2025), which addresses the challenge of missing modalities as a test-time adaptation problem, by incorporating an auxiliary loss during the fine-tuning of our model.

4. Multi-omics joint representation learning

4.1. Bulk RNA-seq and DNA methylation processing for language modeling

Modalities alignment Bulk RNA-seq provides an estimate of the mean expression over all cells in a sample for a large number of genes denoted N_{genes} (typically $N_{genes} \sim 10^4$). Thus, each sample of RNA-seq is composed of real values (in units of transcript per million, or TPM) per gene, $X_{rna} \in \mathbb{R}^{N_{genes}}$, to which we apply a $x \mapsto \log_{10}(1 + x)$ transformation. DNA methylation is the enzymatic attachment of methyl groups to DNA nucleotide bases (usually Cytosine followed by Guanine or CpG site). The methylation level of a given site is expressed by a beta value $\beta \in [0, 1]$, with the number of measured sites, N_{sites} , being around 450,000, resulting in a methylation sample $X_{sites.meth} \in [0, 1]^{N_{sites}}$. The first step in our modeling is to align RNA-seq and methylation data by defining a methylation value per gene $X_{meth} \in \mathbb{R}^{N_{genes}}$. More precisely, for each gene $g \in [1 ; N_{genes}]$, we define $sites(g)$ as all methylation sites associated with the gene, as defined in the Infinium Human Methylation 450k (Bibikova et al., 2011) BeadChip annotations. We evaluate two aggregation strategies for computing gene-level methylation values: (1) a region-weighted strategy where each site $s \in sites(g)$ is assigned weight $w(s)$ based on its genomic region, with higher weights for regulatory regions (Deaton & Bird, 2011), and (2) a simple average where all weights are equal ($w(s) = 1$). Both compute gene-level methylation as $X_{meth}[g] = \frac{\sum_{s \in sites(g)} w(s) \cdot X_{sites.meth}[s]}{\sum_{s \in sites(g)} w(s)}$. Benchmark experiments in the Appendix demonstrate that the simple average already achieves state-of-the-art performance; thus, we use this method by default. Detailed weight specifications and a comprehensive comparison are provided in Appendix D. Finally, a bimodal sample is represented as $X = (X_{rna}, X_{meth}) \in \mathbb{R}^{2N_{genes}}$.

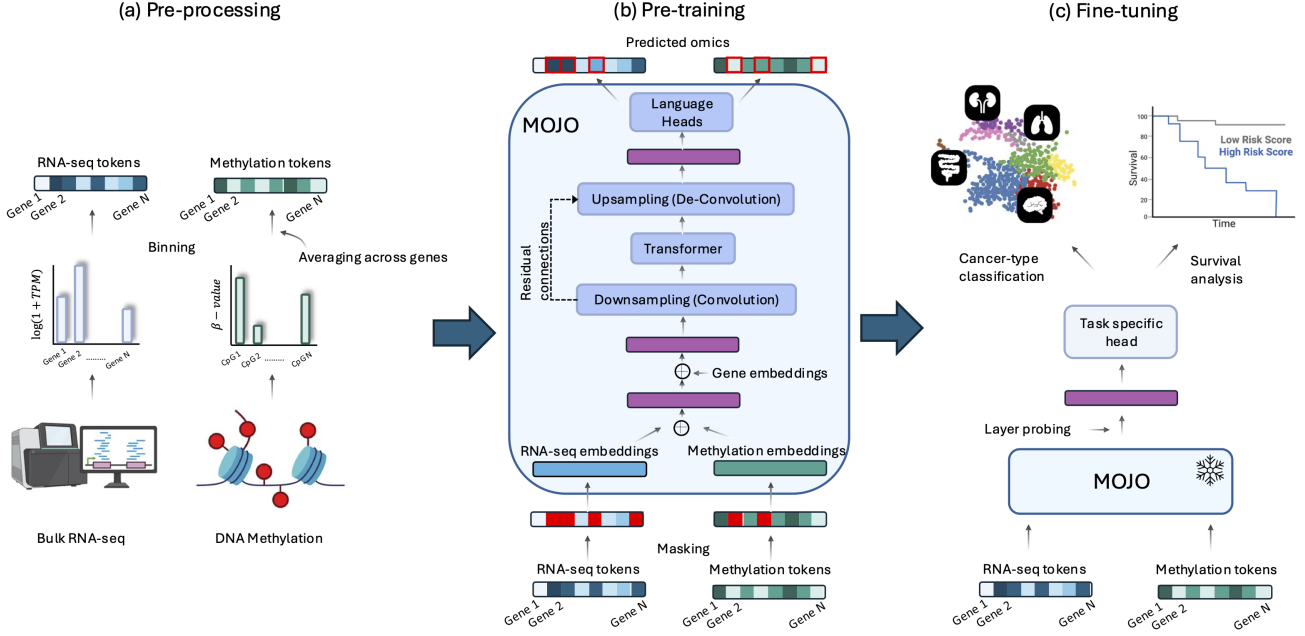


Figure 1. *MOJO* pipeline. (a) Each modality is first tokenized using linear binning. (b) *MOJO*, whose core architecture is composed of a mix of convolution and attention operations, is firstly pre-trained through bimodal masked language modeling. (c) Embeddings are probed from *MOJO* to fine-tune a task-specific head tailored for cancer-type classification or survival analysis.

Tokenization Language models learn to estimate the likelihood of token sequences. Thus, after aligning the two modalities and obtaining a feature vector $X = (X_{rna}, X_{meth})$, each of its components is tokenized by binning their values on linear scales. The token id associated with a given RNA-seq or methylation value is its corresponding bin id. Therefore, after tokenization, one sample is a vector $\tilde{X} = (\tilde{X}_{rna}, \tilde{X}_{meth})$ with $\tilde{X}_{rna} \in \llbracket 0; B_{rna} - 1 \rrbracket^{N_{genes}}$ and $\tilde{X}_{meth} \in \llbracket 0; B_{meth} - 1 \rrbracket^{N_{genes}}$, B_{rna} and B_{meth} being the number of bins respectively for gene expression and methylation. We fix the number of bins for both modalities at 64, and we present an ablation study that evaluates our tokenization strategy in Appendix A.6.

4.2. A long-range model architecture for bimodal representation learning

In order to learn representations of bulk RNA-seq and DNA methylation, we propose a model that combines both convolution and transformer blocks. Inspired by Avsec et al. (2021); Linder et al. (2025); Joshi et al. (2025) to handle long-range genomic dependencies, this architecture allows us to cope with the high dimensionality of the two omics that we consider, each corresponding to a sequence of length N_{genes} . More precisely, a first bimodal embedding is obtained by passing each omic token through classic embedding layers and summing them along with a gene embedding vector. As gene expression and DNA methylation are permutation invariant, this gene embedding acts as a positional

encoding and is randomly initialized. Prior to processing by a transformer model composed of multi-head attention layers (Vaswani et al., 2017), the embedding is downsampled by a convolutional tower. This downscaling allows the transformer block to act on a compressed embedding vector to significantly reduce the computational cost and time. While the convolutional architecture may be counterintuitive for unordered data, it acts as an efficient mechanism for dimensionality reduction. The original sequence length is restored using a deconvolutional tower with residual connections flowing from the downsampling blocks. Separate language modeling heads predict binned gene expressions and methylation. This architecture is summarized in Figure 1 (with a zoom in on the transformer part in Appendix A.2). We also compare *MOJO* to a variant that retains only its transformer component, which we refer to as *MOJO-Transformer*.

4.3. Pre-training: bimodal masked language modeling

Self-supervision loss Our model is pre-trained through self-supervision using multimodal masked language modeling. Although this framework may be applied to more than two modalities that can be processed as a sequence of tokens, we present it in the bimodal case where the set of modalities $\mathbb{M} = \{rna, meth\}$. We adopt standard parameters for masked language modeling (Devlin et al., 2019): for each sequence, 15% of the tokens are corrupted to train the model. Among these corrupted tokens, 80% are replaced with a special $\langle \text{MASK} \rangle$ token, 10% are substituted with random tokens, and the remaining 10% are

left unchanged, but still contribute to the loss. The final heads of our model provide a set of probability distributions $p_m \in [0, 1]^{N_{genes} \times B_m}$ for $m \in \mathbb{M}$. Denoting $\mathcal{M}_m$, for $m \in \mathbb{M}$, the set of masked token indices for that modality, the following multimodal negative log-likelihood is then optimized: $\mathcal{L} = - \sum_{m \in \mathbb{M}} \frac{1}{|\mathcal{M}_m|} \sum_{i \in \mathcal{M}_m} \log((p_m)_{i, (\tilde{x}_m)_i})$

Experiment Our model is pre-trained using the TCGA dataset from 33 cohorts, resulting in 9,252 pairs of bulk RNA-seq and methylation, 5% being kept for testing. We selected $N_{genes} = 17,116$ genes by using the same set of genes as [G  lard et al. \(2025\)](#), from which genes with no methylation data were removed. The model is trained on a total of 192 billion tokens using the Adam ([Kingma & Ba, 2014](#)) optimizer with gradient accumulation to reach an effective batch of 3×10^6 tokens, on a TPU v4-8. The complete set of hyperparameters can be found in Appendix A.1, as well as pre-training learning curves in Appendix A.4.

5. Evaluation downstream tasks

The representations learned by *MOJO* are evaluated using a panel of downstream tasks ranging from supervised classification, survival analysis, and zero-shot classification, to clustering. We compare our method to unimodal (only RNA-seq or DNA methylation) and bimodal models:

BulkRNABert ([G  lard et al., 2025](#)): A transformer-based model pretrained on bulk RNA-seq using masked language modeling. Embeddings are extracted from the last self-attention layer, and the mean embedding over the sequence is used as input for downstream tasks. The tokenization of the RNA-seq data is the same between *MOJO* and *BulkRNABert*, i.e., the same B_{rna} has been used.

scGPT ([Cui et al., 2024](#)): We benchmark *scGPT* by adapting it from its original single-cell context to serve as a bulk RNA-seq encoder. This is achieved by applying the same parameter-efficient fine-tuning procedure used for *BulkRNABert*, effectively repurposing *scGPT* for bulk data across the different tasks.

MethFormer: We develop the equivalent of *BulkRNABert* for DNA methylation (averaged per gene) and we will refer to it in the results as *MethFormer*. Similarly, the same value of B_{meth} is maintained to allow for fair comparison. This model differs from *MethylBERT* ([Jeong et al., 2025](#)) which uses read-level methylome and not the 450k microarrays we are interested in.

Late integration: Bimodal integration resulting from the fusion of embeddings extracted from unimodal models. More precisely, we will refer to *Late integration (concatenation)* as the concatenation of the embeddings from *BulkRNABert* (for RNA-seq) and *MethFormer* (for methylation) which have been pre-trained beforehand. *Late integration (cross-*

attention) corresponds to an integration of the two embeddings with a two-step cross-attention followed by a concatenation, allowing for interaction between the two modalities. The different cross-attention modules are only trained when fitting the downstream tasks. An illustration of the late integration is provided in Appendix B. Finally, we employ the integration strategy from **MultiSurv** ([Vale-Silva & Rohr, 2021](#)), where multimodal representations are aggregated by computing the maximum over features from unimodal models. We apply this aggregation method across all downstream tasks, including those beyond survival analysis, but retain the identifier *MultiSurv* to refer to this specific integration technique.

CustOmics ([Benkirane et al., 2023](#)): A multi-omics model based on Variational Auto-Encoders and tailored for cancer-type classification and survival analysis. Although it can handle up to three modalities (bulk RNA-seq, DNA methylation, and Copy Number Variation), we are here interested in its version that can perform the downstream tasks in the bimodal setting. Two models are considered: *CustOmics (end-to-end)* which trains the VAEs and the task heads jointly, and *CustOmics (probing)* that first learns the unsupervised representation with VAEs and then uses the encoded features as input to task heads.

Multi-Omics Factor Analysis (MOFA) ([Argelaguet et al., 2018](#)): An unsupervised machine learning method designed to integrate multi-omics data by inferring a set of low-dimensional hidden factors, which can be seen as a multi-omics extension to PCA. *MOFA* factors are then used as input either to a Support Vector Machine ([Cortes & Vapnik, 1995](#)) for cancer-type classification or a Cox proportional model ([Cox, 1972](#)) for survival analysis.

SeNMo ([Waqas et al., 2024](#)): A self-normalizing MLP for multi-omics data integration, using ELU activations with *AlphaDropout* ([Klambauer et al., 2017](#)). We adapt *SeNMo* to operate on the concatenation of bulk RNA-seq and DNA methylation features, retrained on the same TCGA splits as *MOJO*.

GAT: Inspired by MOGAT ([Tanvir et al., 2024](#)), this Graph Attention Network builds a k -nearest-neighbour patient-similarity graph over the dataset. Modality-specific linear projections compress RNA-seq and methylation features independently before fusion, and two GAT layers aggregate neighbourhood information to produce patient embeddings. We adapt it to operate on bulk RNA-seq and DNA methylation data exclusively.

An exhaustive benchmark of other feature extraction and multi-omics integration models is included in Appendix C.

Table 1. Cancer type classification. **Bold**: best; underline: second-best.

Model	test weighted-F1
BulkRNABert	0.943 $\pm$ 0.004
scGPT	0.831 $\pm$ 0.008
MethFormer	0.931 $\pm$ 0.006
MOFA	0.852 $\pm$ 0.007
SeNMo	0.922 $\pm$ 0.014
GAT	0.915 $\pm$ 0.004
Late integration (concat.)	0.945 $\pm$ 0.007
Late integration (cross-att.)	0.945 $\pm$ 0.002
MultiSurv	0.932 $\pm$ 0.005

5.1. Cancer-type classification

Methodology *MOJO* is first evaluated on the supervised task of cancer-type classification. The pan-cancer TCGA dataset is divided into 33 cohorts that make up the labels for the classification task. The last attention layer of the transformer component of *MOJO* is probed to obtain the embedding used for classification. After being downsampled by the convolutional layers, the embedding lies in $\mathbf{R}^{n_{\text{downsample}} \times \text{emb}_{\text{dim}}}$, with $\text{emb}_{\text{dim}} = 512$ and $n_{\text{downsample}} = 67$ (resulting from 8 successive downsampling operations by a factor of 2 from an initial sequence length of $N_{\text{genes}} = 17,116$, padded to the next power of 2, 17,152). Then a mean embedding in $\mathbf{R}^{\text{emb}_{\text{dim}}}$ is obtained by averaging over the sequence dimension. This embedding serves as the input to a Multi-Layer Perceptron (MLP) of two hidden layers (respectively of size 256 and 128) that outputs logits for cancer-type prediction. Additionally, we apply dropout (Srivastava et al., 2014) and layer normalization (Ba et al., 2016). *BulkRNABert*, *MethFormer*, and *MOJO* are further fine-tuned in addition to training the MLPs using the parameter-efficient method *IA*³ (Liu et al., 2022), which introduces low-dimensional learnable parameters into the self-attention mechanisms and feed-forward networks. For *MOJO*, we also adapt this principle to the convolutional layers by performing a point-wise multiplication between the output of each convolutional operation and a learnable vector of the same dimension.

Results Cancer-type classification results on the pan-cancer TCGA dataset are presented in Table 1. For this task, the dataset has been split into 80% for training and 20% for testing (stratified by cohort), repeating the split for 5 different seeds. We report the average and standard deviation over these 5 seeds for the different metrics. We will be using both macro F_1 and weighted F_1 scores to avoid any bias due to class imbalance (label distribution is provided in Appendix C.1). *MOJO* provides state-of-the-art results when considering the two modalities, with better performance than its pure Transformer equivalent (*MOJO-Transformer*), *CustOmics* and late integration. For the latter, the *cross-attention* version performs similarly to its *con-*

Model	test weighted-F1
CustOmics (probing)	0.911 $\pm$ 0.088
CustOmics (end-to-end)	<u>0.946 $\pm$ 0.006</u>
MOJO - Transformer (no pre-training)	0.827 $\pm$ 0.010
MOJO - Transformer (probing)	0.892 $\pm$ 0.009
MOJO - Transformer	0.942 $\pm$ 0.003
MOJO (no pre-training)	0.891 $\pm$ 0.006
MOJO (probing)	0.945 $\pm$ 0.006
MOJO	0.952 $\pm$ 0.006

Table 2. Average time per update step (forward + backward pass) during training of classification models on a TPU v4-8. All models are evaluated with an effective batch size of 64, achieved via gradient accumulation when necessary. For each model, we additionally report the maximum batch size supported by the model. As in classification benchmarks, parameter-efficient fine-tuning is applied to *MOJO*, *MOJO - Transformer*, and *BulkRNABert*.

Model	Update time (seconds)	Maximum batch size
BulkRNABert	4.462 $\pm$ 0.006	8
Late integration (cross-attention)	5.819 $\pm$ 0.006	4
Late integration (concatenation)	2.205 $\pm$ 0.004	16
MOJO - Transformer	17.987 $\pm$ 0.001	4
MOJO	0.059 $\pm$ 0.009	1,024

catenation counterpart. Our joint modeling of RNA-seq and methylation with *MOJO* outperforms the corresponding unimodal transformer-based models (*BulkRNABert* and *MethFormer*). Moreover, when only probing the last attention layer and fitting an SVM (*MOJO (probing)* in the table), our model shows a clear performance increase in comparison with *CustOmics (probing)*, highlighting that representations from masked language modeling exhibit stronger predictive capacity. Finally, we show that our bimodal masked language modeling pre-training produces a notable performance gain compared to a model trained from scratch (*MOJO (no pre-training)* in the table).

Training times We additionally report in Table 2 the time required by different models (*BulkRNABert*, *Late integration (cross-attention)*, *Late integration (concatenation)*, *MOJO*, and *MOJO - Transformer*) to perform a full update step (forward and backward pass) when training a pan-cancer classification model. While supporting substantially larger batch sizes compared to purely transformer-based models or late integration mechanisms, *MOJO* achieves approximately a 100 $\times$ speedup over other benchmarked models, and a 300 $\times$ speedup over its transformer-based counterpart. This highlights the computational efficiency of our hybrid architecture that combines convolutional and transformer layers, offering a more scalable alternative to fully transformer-based approaches.

5.2. Survival Analysis

Methodology We then evaluate omics embeddings on a pan-cancer survival task, also known as time-to-event prediction. This task involves predicting the survival time T_i^* for individuals with cancer, specifically the time from diagnosis until death. A key challenge in survival analysis is right censoring, where the observed time C_i^* might be shorter than the actual survival time T_i^* due to factors like the end of a study or loss of patient contact. Consequently, the true target time used by the model, T_i , is defined as the minimum of the actual survival time and the censoring time ($T_i = \min(T_i^*, C_i^*)$). One defines as well $\delta_i = \mathbb{1}_{T_i^* \leq C_i^*}$ (so $\delta_i = 1$ if the event occurred (death), otherwise $\delta_i = 0$), thus constituting a dataset $\mathcal{D} = \{T_i, x_i, \delta_i\}_{i=1}^N$, with x_i the covariates (RNA-seq and/or methylation embeddings in our study). To ensure robust evaluation, data splits for this task were stratified based on the (cohort, event) pair, guaranteeing that each split maintains a consistent distribution of cohorts and observed mortality events. A widely used method to tackle such time-to-event problems is the Cox proportional hazards model (Cox, 1972). This semi-parametric model focuses on modeling the hazard function $\lambda(t|x)$, which represents the instantaneous rate of an event at time t given covariates x . A Cox model expresses λ as $\lambda(t|x) = \lambda_0(t)e^{\hat{h}_\beta(x)}$, with β a vector of parameters (so in our case $x, \beta \in \mathbb{R}^{emb_{dim}}$), $\lambda_0(t)$ the hazard baseline, and in the Cox model, $\hat{h}_\beta(x) = \beta^T x$. More recent works (like *DeepSurv*, or *Cox-nnet* (Katzman et al., 2018; Ching et al., 2018) relax the linear combination of features to replace $\hat{h}_\beta(x)$ with the output of a neural network. For the survival task, we extract the embeddings from *BulkRNA-Bert*, *MethFormer*, and *MOJO* in the same way as for the cancer-type classification task and train a similar MLP architecture on top. We do not consider the cross-attention version of the late integration here as the Cox loss requires working with a sufficiently large batch size, rendering the computation of cross-attentions (given the sequence length of $N_{genes} = 17,116$) computationally prohibitive so that the computation of cross-attentions. Similarly, we do not apply IA^3 and only consider the probing experiment for the survival task (as fine-tuning with IA^3 yielded negligible improvements over probing for this task, see Appendix C.3).

Results Survival results on the pan-cancer TCGA dataset are presented in Table 3. The same split strategy as for the classification task is used. Two different evaluation metrics based on Harrell’s C-index (Harrell et al., 1982) are reported. First, a C-index is computed on the whole test set (all cohorts) referred to as “C-index”. However, in order to make sure that a pan-cancer model is able to predict survival within cohorts correctly, and not just to differentiate survival chances between cancer types, a “Weighted C-index” is also reported. This corresponds to a weighted sum of the C-indexes computed per cohort on the pan-cancer test set, with

Table 3. Pan-cancer survival analysis

Model	C-index	Weighted C-index
BulkRNA-Bert	0.749 $\pm$ 0.003	0.654 $\pm$ 0.014
scGPT	0.720 $\pm$ 0.005	0.604 $\pm$ 0.020
MethFormer	0.736 $\pm$ 0.006	0.622 $\pm$ 0.014
MOFA	0.648 $\pm$ 0.037	0.601 $\pm$ 0.022
CustOmics	0.686 $\pm$ 0.018	0.639 $\pm$ 0.099
Late int. (concat.)	0.756 $\pm$ 0.004	0.653 $\pm$ 0.011
MultiSurv	0.742 $\pm$ 0.002	0.636 $\pm$ 0.011
SeNMo	0.745 $\pm$ 0.004	0.640 $\pm$ 0.017
GAT	0.757 $\pm$ 0.007	0.619 $\pm$ 0.013
MOJO-Transformer	0.757 $\pm$ 0.006	0.657 $\pm$ 0.007
MOJO	0.771 $\pm$ 0.006	0.670 $\pm$ 0.009

weights corresponding to the number of samples of each cohort in the test set. As for the classification task, *MOJO* exhibits higher performance than the unimodal transformer-based models and the late integration. *MOJO* (52M parameters) outperforms *CustOmics* (46M) with a significant gain, confirming that architectural design, specifically the hybrid convolutional-transformer pre-training, rather than model capacity alone drives performance. In an additional experiment, *MOJO*’s performance is matched by an end-to-end version of *CustOmics* (weighted C-index of 0.669 ± 0.004). This need for end-to-end training compared to simple probing highlights the strength of *MOJO*’s learned representations. *MOJO* also outperforms *SeNMo* and *GAT*; notably, the *GAT* model achieves a competitive global C-index but the lowest weighted C-index (0.619), revealing that inter-patient graph comparisons benefit cross-cohort ranking but fail to capture within-cohort survival variation. Kaplan-Meier curves are provided in Appendix C.5, showing better patient stratification with *MOJO*.

5.3. Zero-shot pan-cancer and breast cancer sub-typing and clustering

Methodology To evaluate *MOJO*’s learned embeddings in a fully unsupervised manner, we assess their zero-shot classification and clustering capabilities on PAM50 breast cancer sub-typing (Luminal A, Luminal B, Basal, and HER2) (Parker et al., 2009) and the Pan-cancer dataset from section 5.1. First, zero-shot classification uses a k -nearest neighbors model ($k = 5$), evaluated by accuracy using 5-fold stratified cross-validation, to assess embedding quality without fine-tuning, inspired by Joshi et al. (2025). Second, Leiden clustering (Traag et al., 2019) is performed in the embedding space, with Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) as metrics. We primarily compare the effectiveness of *MOJO*’s joint modeling against late integration (concatenation) embeddings for bimodal data.

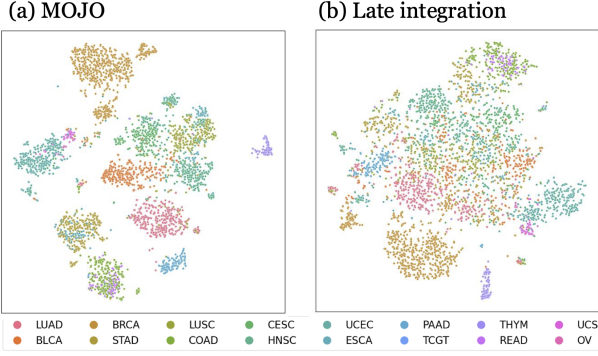


Figure 2. t-SNE representation of *MOJO* and *Late integration* embeddings, colored by cancer-type on a subset of cohorts. *MOJO*’s embeddings visually exhibit better cohort separation capacity compared to *Late Integration* ones, corroborating quantitative results from Table 4. Full pan-cancer t-SNE plot is provided in Appendix C.4.

Results Zero-shot classification and clustering results are shown in Table 4, showing better performance when using *MOJO* embedding than late integration. A comparison with *CustOmics* is also added in Appendix C.2, with lower performance than *MOJO*. Figure 2 presents t-SNE plots of both embeddings in the pan-cancer setting, demonstrating that *MOJO* embeddings achieve superior cohort separation.

Table 4. Zero-shot classification and clustering results on pan-cancer and PAM50 tasks. (Acc. = Accuracy, NMI = Normalized Mutual Information, ARI = Adjusted Rand Index).

Task	Metric	MOJO	Late integration
PAM50	Acc.	0.777	0.763
	NMI	0.345	0.291
	ARI	0.213	0.154
Pan-cancer	Acc.	0.928	0.870
	NMI	0.862	0.771
	ARI	0.756	0.620

5.4. MOJO encodes known biological pathways

While *MOJO*’s representations seem to be powerful for predictive tasks like cancer-type classification and survival analysis, interpreting them is also a critical step. To this end, we focus on a classification model based on *MOJO*’s embeddings (same setting as per Section 5.1) on the PAM50 BRCA sub-typing task (presented in section 5.3). Our classification model maps *MOJO*’s embeddings of $\mathbb{R}^{\text{emb}_{\text{dim}}}$ to logits in $\mathbb{R}^{n_{\text{classes}}}$ with here $n_{\text{classes}} = 4$. We compute Shapley (SHAP) Values (Lundberg & Lee, 2017) for this model and focus on a particular subtype. We identify the most important dimension as the one that maximizes the mean absolute SHAP value. However, SHAP values only inform us about the contributions of the embedding dimensions to the predictions, but without indicating which genes drove the activation of these dimensions. To return to the

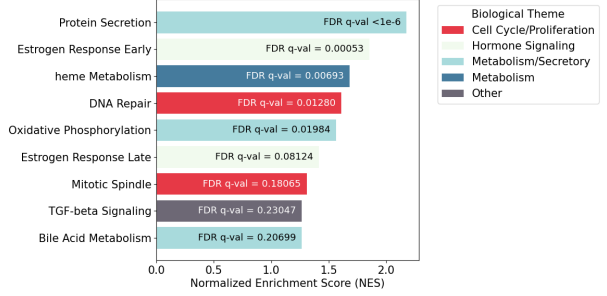


Figure 3. Cancer-relevant enriched pathways for *LuminalB* subtype using pre-rank GSEA from correlations with most informative *MOJO*’s dimensions. Only pathways with positive enrichment scores are represented. (FDR: False Detection Rate).

gene level, for each gene, we compute the Spearman correlation between the value of the embedding at that dimension and the bulk RNA-seq value. This yields a ranked list of genes that is used as input to a pre-ranked Gene Set Enrichment Analysis (GSEA, Subramanian et al. (2005)). We extend this approach to multiple dimensions by considering the average of the Fisher z-transformations of the correlation obtained for the *top k* best dimensions (as per Stouffer’s Z-score method (Stouffer et al., 1949)). Taking the example of *Luminal B* sub-type, Figure 3 shows that known pathways are correctly enriched (like *Protein Secretion* or *Estrogen Response*) (Li et al., 2016; Tran & Bedard, 2011; Fertig et al., 2015), showing that the dimensions that contribute the most to classifying this subtype encode known biological information ($k = 10$ has been used here). Extended results on all BRCA subtypes can be found in Appendix F.

6. External validation and platform generalization

6.1. Cross-platform transferability to legacy 27k arrays

MOJO aggregates CpG probes into gene-level methylation features, making it inherently array-agnostic. To test transferability to legacy 27k platforms, we generated an in-silico 27k dataset by filtering the TCGA 450k data to the 27k microarray probe IDs, maintaining exact patients for a 1-to-1 comparison. Since 27k arrays are heavily promoter-biased, the resulting gene-level values differ substantially from their 450k counterparts (mean shift of -0.115), inducing a severe distribution shift. Under zero-shot evaluation, baseline models collapse: *CustOmics*’ macro- F_1 drops from 0.922 to 0.226 and survival C-index from 0.686 to 0.509. In contrast, zero-shot *MOJO* retains robust performance (C-index: 0.717; macro- F_1 : 0.734), even outperforming a *CustOmics* model fine-tuned on 27k arrays on the survival task. Furthermore, per-gene z-score rescaling (still without any fine-tuning) recovers near full-450k performance (C-index: 0.765; macro- F_1 : 0.909). Detailed methodology and extended metrics are provided in Appendix G.

6.2. Generalization to external held-out cohorts

We assessed *MOJO*’s generalization on two held-out external datasets. On the TARGET cohort, while survival performance is on par with *BulkRNABert* (C-index 0.601 ± 0.023), suggesting methylation provides overlapping rather than additive prognostic information in this pediatric setting, *MOJO*’s bimodal representations demonstrate clear advantages in zero-shot clustering, with substantially higher NMI (0.949 vs. 0.794) and ARI (0.946 vs. 0.706) over late integration. On the ICGC cohort of 405 adult donors across four cancer types, *MOJO* achieves a superior C-index of 0.823 ± 0.019 (Table 5), outperforming both *BulkRNABert* (0.805 ± 0.036) and *MethFormer* (0.810 ± 0.027), confirming that *MOJO*’s bimodal embeddings generalize to entirely unseen cohorts. Complete dataset details and extended evaluation are provided in Appendix E.

Table 5. Survival on the external ICGC cohort.

Model	C-index	Weighted C-index
BulkRNABert	0.805 ± 0.036	0.610 ± 0.087
MethFormer	0.810 ± 0.027	0.628 ± 0.062
MOJO	0.823 ± 0.019	0.643 ± 0.032

7. Missing modalities

While *MOJO*’s joint modeling of two modalities improves downstream task performance, clinical applications may face missing modalities, including the potential for a modality to be entirely unavailable. Being trained by masked language modeling, *MOJO* naturally accepts missing RNA-seq or methylation information for a given subset of genes by attributing a special <MASK> token to these genes. One can naturally extend this procedure by assigning a sequence entirely composed of <MASK> tokens to a missing modality, making *MOJO* inherently capable of handling missing modalities. However, as during pre-training only a fraction of each modality is masked to account for the masked language modeling loss, the absence of a whole modality is never encountered by the model. To this end, we conducted an additional pre-training of *MOJO* by incorporating samples from the TCGA dataset that are missing one of the two considered modalities, thus extending the initial pre-training dataset composed of 9,252 pairs (X_{rna}, X_{meth}) with 2,022 pairs ($X_{rna}, None$) and 560 pairs ($None, X_{meth}$) with *None* indicating a missing modality. We will refer to this model as *MOJO-Extended-Pretraining*. Two settings thus arise: (S1) one modality is either missing during model fine-tuning and we want to assess the utility of *MOJO*’s embeddings in this situation, or (S2) it is missing only at test-time while the downstream model has been trained with full bimodal input.

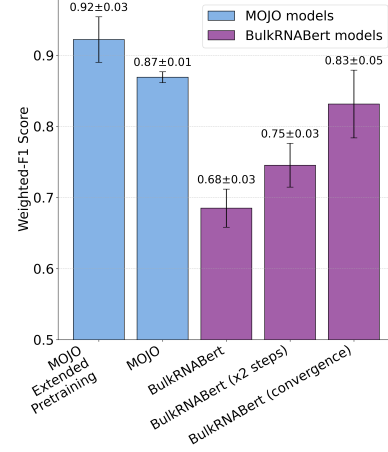


Figure 4. Ovarian cancer sub-typing: *MOJO* outperforms *BulkRNABert* while being faster to fine-tune. (“*BulkRNABert* models” bars from left to right: same fine-tuning budget as *MOJO*, $\times 2$ fine-tuning steps, and until convergence). *MOJO-Extended-Pretraining* refers to an extension of the pre-training dataset with samples with missing modalities.

7.1. Missing modalities for fine-tuning (S1)

In the context of Ovarian (OV) cancer subtyping in TCGA (4 classes: differentiated, immunoreactive, mesenchymal, and proliferative (Verhaak et al., 2012)), only RNA-seq samples are available. A bimodal pre-trained *MOJO* model is thus fine-tuned on this task with ($X_{rna}, None$) as input and achieves superior performance than *BulkRNABert* while being faster to train (Figure 4). Using *MOJO-Extended-Pretraining* instead of *MOJO* further improves the performance in OV subtyping. In this context, we prove that *MOJO*, while initially being a bimodal model, can be used as an embedding model for RNA-seq only, and provide better performance than a specific unimodal model.

7.2. Missing modalities at test-time: Mutual Information loss (S2)

Motivation Our goal is to create bimodal models for tasks like the pan-cancer classification task (for which both modalities are available during fine-tuning) that maintain performance comparable to unimodal models when a modality is absent at test time. Our framework begins with training a bimodal downstream model fine-tuned using complete (X_{rna}, X_{meth}) pairs from a pre-trained *MOJO* (per section 5.1). Our contribution involves modifying this fine-tuning for missing modalities.

Methodology At test time, we compute metrics under two settings: with the complete test set, and by simulating the absence of either RNA-seq or methylation by dropping it from $x\%$ of test pairs (up to $x=100\%$ for removal in all samples), all without further model fine-tuning. We only fine-tune and evaluate the downstream model with samples that have both modalities, thus allowing us to get a fair comparison between all the models. Since the dataset is

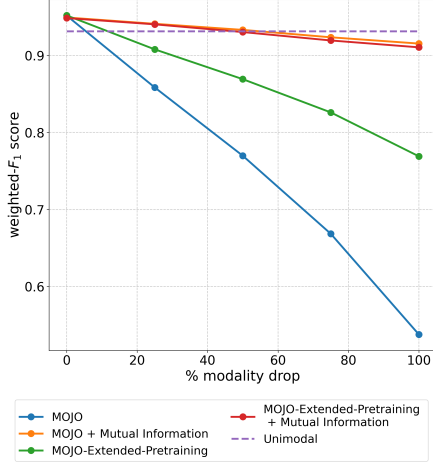


Figure 5. Performance when dropping RNA-seq. Test weighted-F1 score is reported as a function of the percentage of dropped RNA-seq samples in the test set. “+ Mutual Information” indicates the use of MI auxiliary loss during fine-tuning.

fixed, we must modify the fine-tuning procedure to maintain performance despite the complete absence of a modality at test time. To this end, we add mutual information as an auxiliary loss paired with classic cross-entropy for the pan-cancer classification task. This quantity is used in Ramazanov et al. (2025) as a test-time adaptation technique of an audio/vision model to handle missing modalities. We adapt it to be directly incorporated during the fine-tuning phase of the model to avoid any modification of the model at test time, thus saving computational time. Following the notation from Ramazanov et al. (2025), we denote $f_\theta(x; m)$ the output of the classification model for a given input x when modalities m are present, with $m \in \mathcal{D}_{modality} = \{rna + meth, rna, meth\}$. One would require f_θ to provide the same prediction regardless of the provided modality, thus satisfying the following equality: $f_\theta(x; rna + meth) = f_\theta(x; rna) = f_\theta(x; meth)$. One can satisfy such a constraint by minimizing the following loss: $\mathcal{L}_{aux} = \mathbb{E}_{m \in \mathcal{D}_{modality}} [MI(f_\theta(x, m), m)]$, with $MI(X, Y) = D_{KL}[P_{(X, Y)} || P_X \otimes P_Y]$ corresponding to the mutual information (Shannon, 1948) between two random variables X and Y . The mutual information is equal to 0 when the two random variables are independent; thus, minimizing this quantity as an auxiliary loss should guide the model towards providing the same output independently of the given modality. Therefore, the following loss is now optimized: $\mathcal{L} = \mathcal{L}_{task} + \lambda * \mathcal{L}_{aux}$, with $\mathcal{L}_{task}$ corresponding here to the cross-entropy for the cancer-type classification task. We detail in the algorithm of Appendix H the procedure used to compute this loss for a single example.

Results Results from the missing modalities experiments are shown in Figure 5 when RNA-seq is dropped (similar results when dropping DNA methylation in Appendix H). The initial *MOJO* model achieved a test weighted- F_1 of 0.952. Dropping all RNA-seq from the test set ($x = 100\%$)

significantly decreases performance to 0.538. Adding mutual information during fine-tuning (we used $\lambda = 10$ as the ratio of the cross-entropy and the mutual information computed after model initialization, see Appendix H.3 for an ablation study on this hyperparameter) maintained stable performance (0.949) with no modality drop. However, it corrected the performance decrease when RNA-seq was missing (from 0.538 to 0.916). These recovered scores approach those of unimodal models (i.e. *MethFormer* as methylation is the remaining modality). Extending pre-training with pairs missing one modality (*MOJO-Extended-Pretraining*) also narrowed the performance gap without the mutual information loss, and combining *MOJO-Extended-Pretraining* with the auxiliary mutual information loss yielded performance similar to using the auxiliary loss alone.

8. Conclusion

We introduced *MOJO*, a novel methodology for learning joint representations from multi-omics data, specifically bulk RNA-seq and DNA methylation. Pre-trained using self-supervised bimodal masked language modeling, *MOJO*’s architecture, combining convolutional and attention components, efficiently handles long gene sequences, outperforming purely transformer-based unimodal models. *MOJO*’s learned embeddings achieve state-of-the-art performance in supervised tasks like cancer-typing and time-to-event prediction on the TCGA dataset compared to unimodal models. In particular, the predictive capacity of *MOJO*’s representations is highlighted by the significant performance gain observed in the layer probing setup. Joint modeling’s advantage over late integration was demonstrated through zero-shot classification and clustering. Although *MOJO* inherently accepts missing modalities, we narrowed the performance gap with unimodal models which arises when a modality is missing by incorporating a mutual information-based loss during fine-tuning. Future work includes extending *MOJO* to more data types, relaxing modality alignment requirements, extending the mutual information approach to more modalities, and improving it to exactly match unimodal performance.

Limitations This work presents a robust framework for multi-omics integration and missing modality handling, improving the potential clinical utility of deep learning-based cancer prognosis. However, some limitations remain. First, *MOJO*’s gene-level methylation averaging acts as an information bottleneck, discarding potentially site-level biological variations. Second, while pre-trained on the entirety of the paired TCGA cohort, this dataset is small relative to other modality foundation models and inherently reflects TCGA’s specific population demographic biases. Deploying such prognostic models requires extensive prospective validation, as relying on these biases may lead to skewed clinical predictions across different populations.

Impact Statement

This paper presents work towards an integration of multi-omics data while handling missing modalities. This approach improves the clinical utility of deep learning-based cancer prognosis by enabling more reliable patient stratification.

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A. MOJO pre-training

A.1. Hyperparameters

Table 6. MOJO model and pre-training hyperparameters

Model Hyperparameters	
Number of downsamples	8
Kernel size	5
Embedding dimension	512
Number of transformer layers	8
Feed forward dimension	1,024
Number of attention heads	16
Training Hyperparameters	
Batch size	128
Gradient accumulation	4
Learning rate	5×10^{-5}
Masking ratio	15%

A.2. MOJO Transformer block

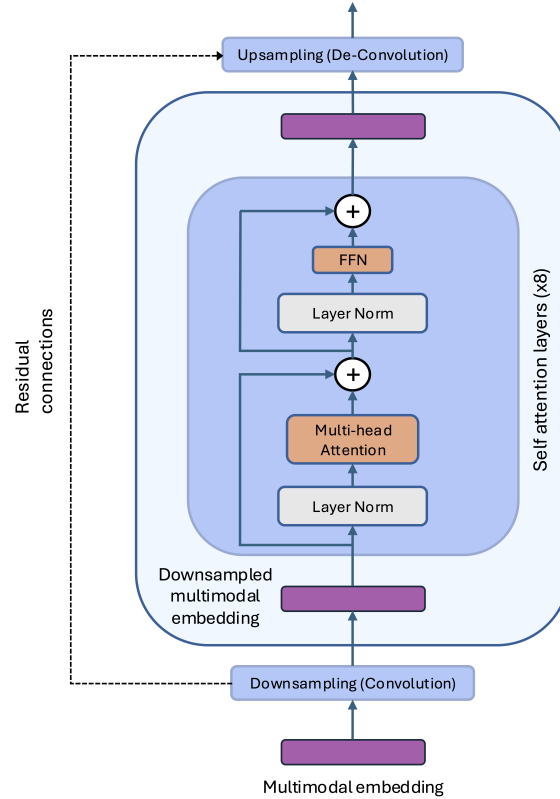


Figure 6. Zoom on the Transformer block of the MOJO architecture. Full architecture is provided in Figure 1.

A.3. Pre-training dataset statistics and preprocessing

Table 7. Composition of the MOJO pre-training datasets.

Model	Split	# Samples	Modality	Cohorts
MOJO	Train	8,789 (95%)	Bimodal (RNA-seq + Methylation)	33 TCGA cohorts
	Test	463 (5%)	Bimodal (RNA-seq + Methylation)	
	Total	9,252		
MOJO-EXTENDED	Train	11,242 (95%)	Bimodal + RNA-seq only + Methylation only	33 TCGA cohorts
	Test	592 (5%)	Bimodal + RNA-seq only + Methylation only	
	Total	11,834		

Preprocessing pipeline. RNA-seq values (TPM) were transformed as $x \mapsto \log_{10}(1 + x)$. For methylation, each sample’s 450k beta-values were mapped to gene-level averages by averaging all CpG sites annotated to that gene in the Illumina HumanMethylation450k BeadChip manifest (simple unweighted average, see Appendix D for the region-aware alternative). We retain $N_{genes} = 17,116$ genes present in both the RNA-seq and 450k annotations. Both modalities were tokenized using linear binning into 64 bins (RNA-seq: $[0, \log_{10}(1 + \max)]$; methylation: $[0, 1]$). The vocabulary comprises 64×2 omic tokens plus special tokens (`<MASK>`, `<PAD>`).

A.4. Pre-training learning curves

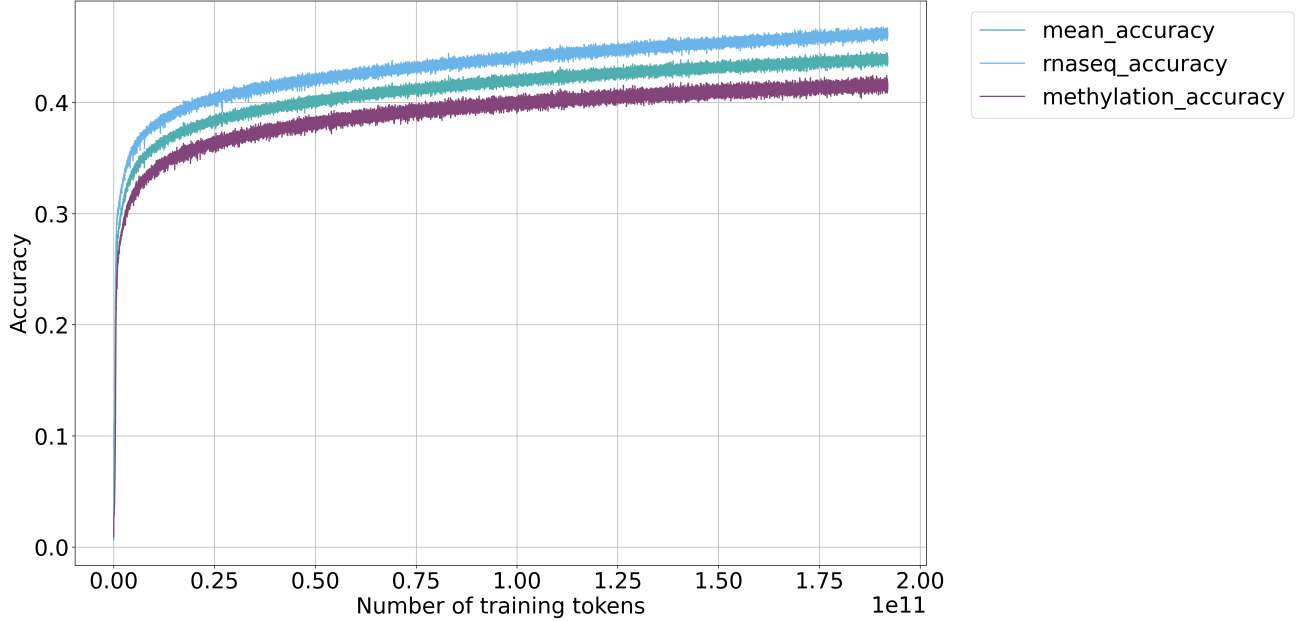


Figure 7. Bimodal masked language modeling pre-training curves of the *MOJO* architecture. The training reconstruction accuracy is represented for each omic separately as well as the average reconstruction accuracy among the different omics.

A.5. Architectural justification: convolutions and input ordering

A potential concern regarding the application of convolution-based architectures, such as the U-Net, to bulk omics data is the lack of inherent spatial locality in the input. Unlike images, where pixel adjacency correlates with semantic structure, genes or CpG sites in a feature vector do not possess a natural grid-like order. However, our adoption of the U-Net backbone is motivated by computational efficiency and multi-scale representation rather than spatial inductive biases. Specifically, the U-Net facilitates hierarchical dimensionality reduction, essential for handling high-dimensional inputs (hundreds of thousands of features), and enables the aggregation of signals ranging from individual gene values to broad molecular patterns.

To empirically validate that the model’s performance does not rely on arbitrary input adjacency, we evaluated a variant, MOJO-ORDERED, where input features were sorted by genomic coordinates (chromosome and physical position). This ordering imposes a biological locality that a convolutional network could theoretically exploit.

Table 8. Impact of genomic ordering: performance comparison between the standard unordered MOJO and the genomically sorted MOJO-ORDERED variant. Results indicate that imposing biological order yields no statistically significant improvement, suggesting the architecture is effectively order-invariant in this context.

Model	Cancer-type Classification		Survival Analysis	
	Macro-F1	Weighted-F1	C-index	Weighted C-index
MOJO (Unordered)	0.935 $\pm$ 0.007	0.952 $\pm$ 0.005	0.771 $\pm$ 0.006	0.670 $\pm$ 0.009
MOJO-ORDERED	0.936 $\pm$ 0.006	0.954 $\pm$ 0.002	0.775 $\pm$ 0.007	0.672 $\pm$ 0.019

As shown in Table 8, MOJO-ORDERED performs comparably to the standard model across both classification and survival tasks, with differences falling within the margin of error. These results confirm that the architectural benefits of MOJO stem from its capacity for hierarchical feature extraction in high-dimensional spaces, rather than local spatial correlations. Consequently, we retain the unordered design for its generality and simplicity.

A.6. Tokenization sensitivity analysis

We analyzed the discretization strategy used to tokenize continuous bulk RNA-seq and DNA methylation values. All models were pre-trained from scratch with the corresponding binning scheme and fine-tuned on the same downstream tasks. Two strategies are compared: *linear* tokenization, which divides the value range into equally spaced bins, and *quantile* tokenization, which sets bin boundaries according to each gene’s empirical distribution so that each bin contains approximately the same number of samples.

Table 9. Sensitivity of MOJO to tokenization strategy and bin granularity. **Bold**: default configuration.

Strategy	Bins	Cancer-type Classification		Survival Analysis	
		Macro-F1	Weighted-F1	C-index	Weighted C-index
Linear	5	0.910 $\pm$ 0.007	0.937 $\pm$ 0.005	0.758 $\pm$ 0.009	0.650 $\pm$ 0.020
Linear	16	0.925 $\pm$ 0.005	0.948 $\pm$ 0.004	0.770 $\pm$ 0.007	0.669 $\pm$ 0.013
Linear	32	0.937 $\pm$ 0.006	0.953 $\pm$ 0.004	0.772 $\pm$ 0.005	0.674 $\pm$ 0.020
Linear	64 (default)	0.935 $\pm$ 0.007	0.952 $\pm$ 0.006	0.771 $\pm$ 0.006	0.670 $\pm$ 0.009
Linear	128	0.941 $\pm$ 0.006	0.954 $\pm$ 0.003	0.770 $\pm$ 0.007	0.664 $\pm$ 0.021
Quantile	64	0.934 $\pm$ 0.008	0.952 $\pm$ 0.006	0.771 $\pm$ 0.006	0.670 $\pm$ 0.008

Performance is stable for ≥ 32 bins, with 64 bins providing the best trade-off between capturing fine-grained biological variation and filtering technical noise. The coarser 5-bin scheme degrades macro- F_1 by 2.7 points and C-index by 1.7 points, confirming that sufficient resolution is important for discriminating subtle expression states. Quantile-based tokenization achieves results equivalent to linear scaling, validating our default tokenization choice.

B. Late integration

An illustration of the late integration mechanism is provided in Figure 8.

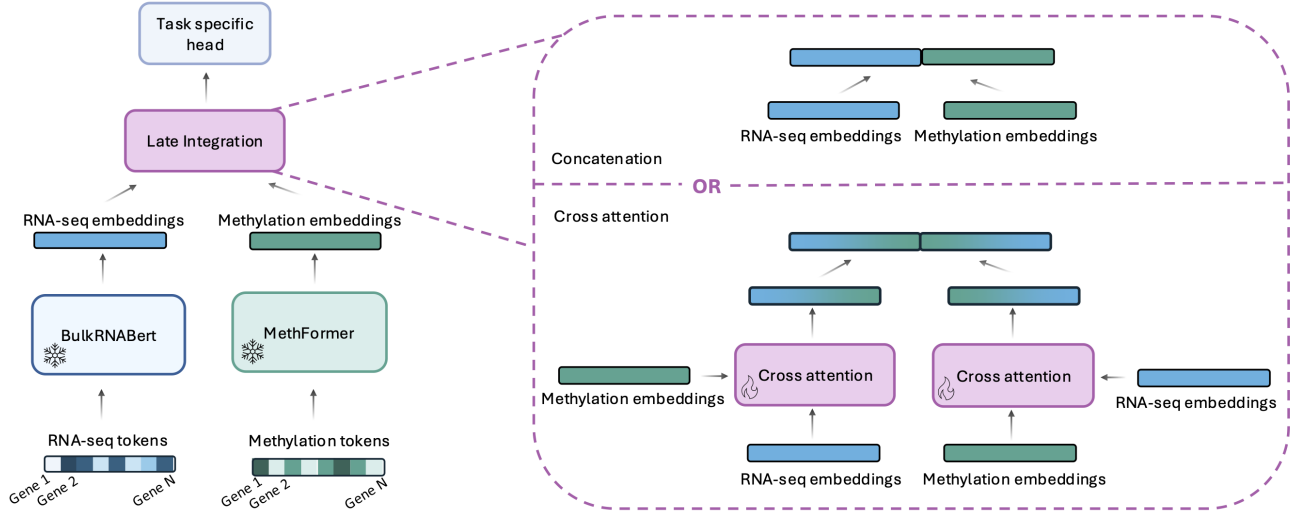


Figure 8. Late integration architecture. RNA-seq and methylation embeddings are obtained from pre-trained transformer based encoders (respectively *BulkRNABert* and *MethFormer*) and are fused either by concatenation or by a two-step cross-attention mechanism.

C. Downstream tasks dataset and benchmarks

C.1. Pan-cancer classification dataset

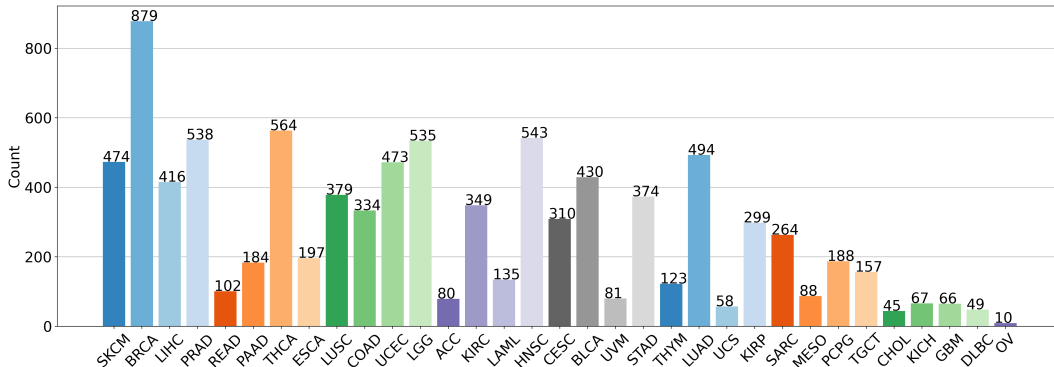


Figure 9. Pan-cancer classification label distribution.

C.2. Downstream tasks benchmarks

In addition to Table 1 (cancer-type classification) and Table 3 (survival analysis), an exhaustive benchmark including other representation models for RNA-seq and DNA methylation was conducted:

- Multiple Factor Analysis (MFA) (Sánchez et al., 2012), using a latent space of dimension 256.
- Non-negative Matrix Factorization (NMF) (Lee & Seung, 1999), with the same latent space dimension as for MFA.
- *OmiEmbed* (Zhang et al., 2021b): a unified multi-task deep learning framework for multi-omics data based on Variational Auto-Encoders (Kingma & Welling, 2013) from early integrated omics.

- *IntegrAO* (Ma et al., 2024): an unsupervised framework based on Graph Neural Networks (Scarselli et al., 2008) for integrating incomplete multi-omics data, tailored for classification and survival task.

Multiple Factor Analysis and Non-negative Matrix Factorization features are then fed to a Support Vector Machine (SVM) for the cancer-type classification task and to a Cox proportional hazards model for the survival analysis task. The results are presented in Table 10 and Table 11. We also report in Table 12 the complete benchmark on the zero-shot classification and clustering tasks which includes *CustOmics* model.

Table 10. Full benchmark on cancer-type classification. **Bold**: best; underline: second-best.

Model	Modality	Macro-F1	Weighted-F1
<i>Unimodal encoders</i>			
BulkRNABert	RNA-seq	0.918 ± 0.008	0.943 ± 0.004
scGPT	RNA-seq	0.793 ± 0.009	0.831 ± 0.008
MethFormer	Methylation	0.917 ± 0.008	0.931 ± 0.006
<i>Classical multi-omics integration</i>			
MFA	Bimodal	0.753 ± 0.013	0.848 ± 0.008
NMF	Bimodal	0.725 ± 0.011	0.827 ± 0.006
MOFA	Bimodal	0.789 ± 0.012	0.852 ± 0.007
<i>Deep learning multi-omics integration</i>			
IntegrAO	Bimodal	0.912 ± 0.005	0.911 ± 0.015
OmiEmbed	Bimodal	0.919 ± 0.004	0.922 ± 0.016
CustOmics (probing)	Bimodal	0.887 ± 0.065	0.911 ± 0.088
CustOmics (end-to-end)	Bimodal	0.922 ± 0.006	<u>0.946 ± 0.006</u>
SeNMo	Bimodal	0.900 ± 0.013	0.922 ± 0.014
GAT	Bimodal	0.882 ± 0.008	0.915 ± 0.004
<i>Late integration (pre-trained encoders)</i>			
MultiSurv	Bimodal	0.911 ± 0.005	0.932 ± 0.005
Late integration (concatenation)	Bimodal	0.928 ± 0.008	0.945 ± 0.007
Late integration (cross-attention)	Bimodal	<u>0.929 ± 0.005</u>	0.945 ± 0.002
<i>Joint bimodal pre-training (MOJO variants)</i>			
MOJO-Transformer (no pre-training)	Bimodal	0.802 ± 0.008	0.827 ± 0.010
MOJO-Transformer (probing)	Bimodal	0.850 ± 0.014	0.892 ± 0.009
MOJO-Transformer	Bimodal	0.925 ± 0.005	0.942 ± 0.003
MOJO (no pre-training)	Bimodal	0.835 ± 0.015	0.891 ± 0.006
MOJO (probing)	Bimodal	0.928 ± 0.009	0.945 ± 0.006
MOJO	Bimodal	0.935 ± 0.007	0.952 ± 0.006

Table 11. Full benchmark on pan-cancer survival analysis. **Bold**: best; underline: second-best.

Model	Modality	C-index	Weighted C-index
<i>Unimodal encoders</i>			
BulkRNABert	RNA-seq	0.750 ± 0.004	0.657 ± 0.011
scGPT	RNA-seq	0.720 ± 0.005	0.604 ± 0.020
MethFormer	Methylation	0.735 ± 0.006	0.618 ± 0.017
<i>Classical multi-omics integration</i>			
MFA	Bimodal	0.616 ± 0.033	0.593 ± 0.016
NMF	Bimodal	0.616 ± 0.040	0.591 ± 0.025
MOFA	Bimodal	0.648 ± 0.037	0.601 ± 0.022
<i>Deep learning multi-omics integration</i>			
IntegrAO	Bimodal	0.710 ± 0.008	0.624 ± 0.006
OmiEmbed	Bimodal	0.736 ± 0.006	0.631 ± 0.007
CustOmics	Bimodal	0.686 ± 0.018	0.639 ± 0.099
SeNMos	Bimodal	0.745 ± 0.004	<u>0.640 ± 0.017</u>
GAT	Bimodal	<u>0.757 ± 0.007</u>	0.619 ± 0.013
<i>Late integration (pre-trained encoders)</i>			
Late integration (concat.)	Bimodal	0.756 ± 0.004	0.653 ± 0.011
MultiSurv	Bimodal	0.742 ± 0.002	0.636 ± 0.011
<i>Joint bimodal pre-training (MOJO variants)</i>			
MOJO-Transformer	Bimodal	0.757 ± 0.006	0.657 ± 0.007
MOJO	Bimodal	0.771 ± 0.006	0.670 ± 0.009

Table 12. Full benchmark on zero-shot classification and clustering results on pan-cancer and PAM50 tasks. (Acc. = Accuracy, NMI = Normalized Mutual Information, ARI = Adjusted Rand Index).

Task	Metric	MOJO	Late integration	CustOmics
PAM50	Acc.	0.777	0.763	0.765
	NMI	0.345	0.291	0.311
	ARI	0.213	0.154	0.176
Pan-cancer	Acc.	0.928	0.870	0.905
	NMI	0.862	0.771	0.830
	ARI	0.756	0.620	0.699

C.3. IA^3 fine-tuning for Survival Analysis

For classification, we apply IA^3 fine-tuning to *MOJO*, *BulkRNABert*, and *MethFormer*. For survival analysis, we use probing only (frozen encoders) due to batch size requirements. To validate that this choice does not disadvantage *MOJO*, we compare probing vs. IA^3 fine-tuning for survival in Table 13.

Table 13. Effect of IA^3 fine-tuning on survival analysis. Differences are within the margin of error, confirming that probing is sufficient for this task.

Model	IA^3	C-index	Weighted C-index
MOJO	✗	0.771 ± 0.006	0.670 ± 0.009
MOJO	✓	0.771 ± 0.007	0.670 ± 0.010
BulkRNABert	✗	0.749 ± 0.003	0.654 ± 0.014
BulkRNABert	✓	0.752 ± 0.004	0.657 ± 0.010
MethFormer	✗	0.736 ± 0.006	0.622 ± 0.014
MethFormer	✓	0.737 ± 0.006	0.620 ± 0.016

C.4. Bimodal embeddings t-SNE visualisations

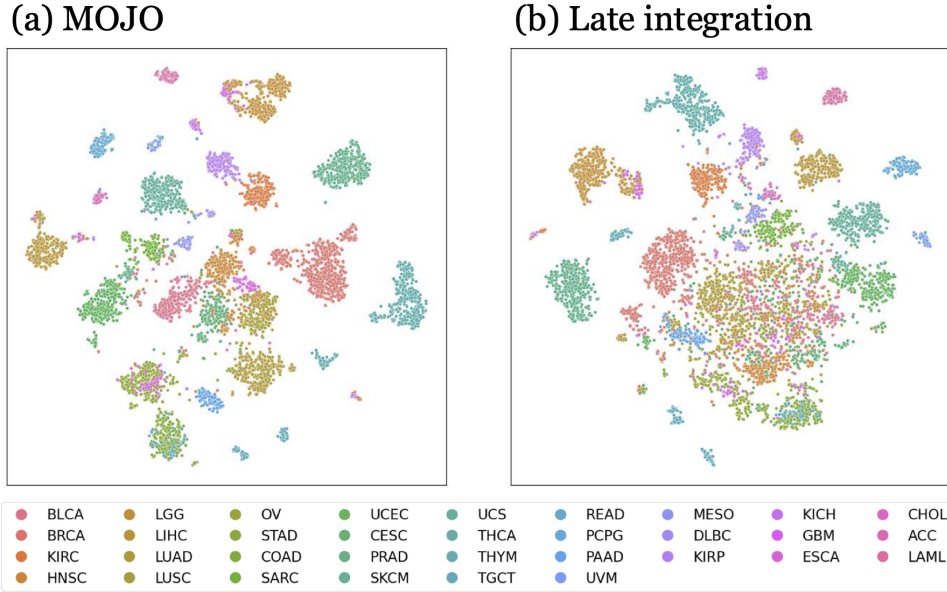


Figure 10. Pan-cancer version of the t-SNE representation of *MOJO* and *Late integration* embeddings, colored by cancer-type.

C.5. Kaplan-Meier curves

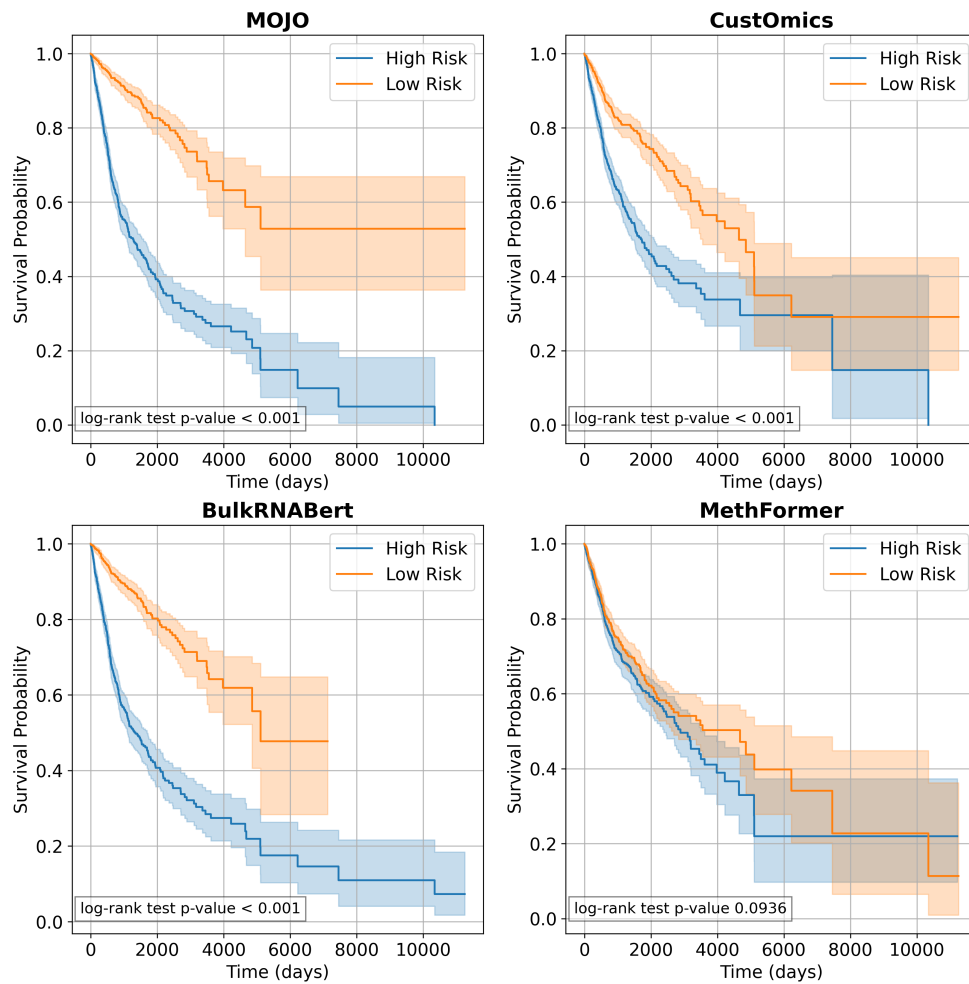


Figure 11. Kaplan-Meier curve for pan-cancer survival models for four models: *MOJO*, *CustOmics*, *BulkRNABert*, *MethFormer*. *MOJO* shows better patient stratification. “Low Risk” and “High Risk” are defined from the median of the log risk computed on the test set for each model.

D. Region-Aware Methylation Aggregation

D.1. Methodology

To map site-level methylation data to gene-level features compatible with bulk RNA-seq, in addition to the simple average of methylation values used throughout our work, we implemented a region-aware weighted aggregation strategy that accounts for the functional heterogeneity of genomic regions. We assigned varying weights to CpG sites based on their genomic location, reflecting the empirically observed distance-dependent impact of methylation on gene expression. Promoter-proximal regions (TSS200, TSS1500) received the highest weights (3.0, 2.5) due to their strong negative correlation with gene expression and established role in transcriptional silencing (Weber et al., 2007; Jones & Baylin, 2007), while 5'UTR and first exon sites received intermediate weights (2.0) based on their impact on transcriptional silencing and inhibition of elongation (Brenet et al., 2011). Gene body and 3'UTR regions served as baseline (weight = 1.0) given their weaker and often positive correlation with transcription (Ball et al., 2009), creating a biologically grounded weighting scheme that prioritizes functionally critical regulatory sites.

Additionally, we incorporated CpG island context weights to capture the differential regulatory potential of CpG sites based on their proximity to CpG islands. CpG islands (weight = 2.0) are strongly associated with gene regulation and promoter activity (Deaton & Bird, 2011), while CpG shores (weight = 1.5) have been shown to exhibit tissue-specific differential methylation with functional consequences (Irizarry et al., 2009). CpG shelves (weight = 1.0) and open sea regions (weight = 0.8) received lower weights reflecting their more variable and less consistent associations with gene expression.

The final weight for each CpG site s associated with gene g was computed as:

$$w(s) = w_{region}(s) \times w_{island}(s)$$

where $w_{region}(s)$ is the genomic region weight and $w_{island}(s)$ is the CpG island context weight, as detailed in Table 14.

Table 14. Weights assigned to CpG sites based on genomic region and CpG island context for region-aware methylation aggregation.

Genomic Region	w_{region}	CpG Island Context	w_{island}
TSS200	3.0	Island	2.0
TSS1500	2.5	Shore	1.5
5'UTR	2.0	Shelf	1.0
1st Exon	2.0	Open Sea	0.8
Gene Body	1.0		
3'UTR	1.0		

The weighted methylation value for each gene was then computed as:

$$X_{meth}[g] = \frac{\sum_{s \in sites(g)} w(s) \cdot X_{sites.meth}[s]}{\sum_{s \in sites(g)} w(s)}$$

where $sites(g)$ represents all CpG sites mapped to gene g according to the Illumina HumanMethylation450 BeadChip annotations (Bibikova et al., 2011; Sandoval et al., 2011), and $X_{sites.meth}[s]$ is the beta value for site s .

D.2. Ablation study

We benchmark the weighted average of methylation value against the simple average on the pancancer cancer-type classification (Table 15) and survival analysis (Table 16) tasks.

Table 15. Ablation study on weighted average of methylation beta values for cancer type classification

Non-weighted average of methylation beta value		Weighted average of methylation beta value	
Model	Test Weighted-F1	Model	Test Weighted-F1
BulkRNABert*	0.943 $\pm$ 0.004	BulkRNABert*	—
scGPT*	0.831 $\pm$ 0.008	scGPT*	—
MethFormer	0.931 $\pm$ 0.006	MethFormer	0.935 $\pm$ 0.005
MFA	0.848 $\pm$ 0.008	MFA	0.850 $\pm$ 0.009
NMF	0.827 $\pm$ 0.006	NMF	0.824 $\pm$ 0.006
MOFA	0.852 $\pm$ 0.007	MOFA	0.849 $\pm$ 0.008
IntegrAO	0.911 $\pm$ 0.015	IntegrAO	0.944 $\pm$ 0.015
OmiEmbed	0.922 $\pm$ 0.016	OmiEmbed	0.925 $\pm$ 0.017
Late integration (concat.)	0.945 $\pm$ 0.007	Late int. (concat.)	0.940 $\pm$ 0.005
Late integration (cross-att.)	0.945 $\pm$ 0.002	Late int. (cross-att.)	0.941 $\pm$ 0.004
Late integration (max.)	0.911 $\pm$ 0.005	Late int. (max.)	0.930 $\pm$ 0.010
CustOmics (probing)	0.911 $\pm$ 0.088	CustOmics (probing)	0.893 $\pm$ 0.071
CustOmics (e2e**)	0.946 $\pm$ 0.006	CustOmics (e2e**)	0.941 $\pm$ 0.003
SeNMo	0.922 $\pm$ 0.014	SeNMo	0.937 $\pm$ 0.006
GAT	0.915 $\pm$ 0.004	GAT	0.920 $\pm$ 0.004
MOJO-Transformer (probing)	0.892 $\pm$ 0.009	MOJO-Transformer (probing)	0.896 $\pm$ 0.007
MOJO-Transformer	0.942 $\pm$ 0.003	MOJO-Transformer	0.938 $\pm$ 0.010
MOJO (probing)	0.945 $\pm$ 0.006	MOJO (probing)	0.950 $\pm$ 0.006
MOJO	0.952 $\pm$ 0.006	MOJO	0.951 $\pm$ 0.005

*BulkRNABert and scGPT use only RNA data and are unaffected by methylation preprocessing. **e2e = end-to-end

Table 16. Ablation study on weighted average of methylation beta values for survival analysis

Non-weighted average of methylation beta value		Weighted average of methylation beta value	
Model	Weighted C-index	Model	Weighted C-index
BulkRNABert*	0.657 ± 0.011	BulkRNABert*	—
scGPT*	0.604 ± 0.020	scGPT*	—
MethFormer	0.618 ± 0.017	MethFormer	0.622 ± 0.018
MFA	0.593 ± 0.016	MFA	0.594 ± 0.018
NMF	0.591 ± 0.025	NMF	0.590 ± 0.026
MOFA	0.601 ± 0.022	MOFA	0.598 ± 0.024
IntegrAO	0.624 ± 0.006	IntegrAO	0.625 ± 0.006
OmiEmbed	0.631 ± 0.007	OmiEmbed	0.633 ± 0.007
CustOmics	0.639 ± 0.099	CustOmics	0.642 ± 0.023
Late integration (concat.)	0.653 ± 0.011	Late integration (concat.)	0.652 ± 0.016
MultiSurv	0.636 ± 0.011	MultiSurv	0.635 ± 0.021
SeNMo	0.640 ± 0.017	SeNMo	0.644 ± 0.021
GAT	0.619 ± 0.013	GAT	0.615 ± 0.018
MOJO-Transformer	0.657 ± 0.007	MOJO-Transformer	0.641 ± 0.010
MOJO	0.670 ± 0.009	MOJO	0.654 ± 0.018

*BulkRNABert and scGPT use only RNA data and are unaffected by methylation preprocessing.

While the region-aware weighted methylation aggregation strategy was designed to better capture the differential regulatory potential of CpG sites, our empirical evaluation reveals nuanced results. As shown in Tables 15 and 16, this biologically motivated weighting scheme does not yield improvements for the MOJO model in either cancer-type classification or survival prediction tasks; therefore, we retain the simpler average-based aggregation in our final model. This confirms that *MOJO* learns the relevant regulatory structure directly from the data, obviating the need for hand-crafted methylation priors. However, when examining the *MethFormer* baseline, we observe marginal improvements with region-aware aggregation. Furthermore, during pre-training, the region-aware scheme achieves consistently higher reconstruction accuracy compared to simple averaging (Figure 12). This improved reconstruction performance suggests that weighted aggregation, by accounting for the differential regulatory roles of promoters, gene bodies, and CpG islands, yields a more biologically grounded representation of methylation patterns. Nevertheless, these improvements do not transfer to the multimodal setting, indicating that the additional biological signal captured by region-aware weighting may be redundant with information already encoded within the RNA-seq modality. This finding highlights a broader challenge in multimodal learning: biologically motivated features that improve unimodal representation quality do not necessarily enhance multimodal fusion when modalities contain overlapping biological signals.

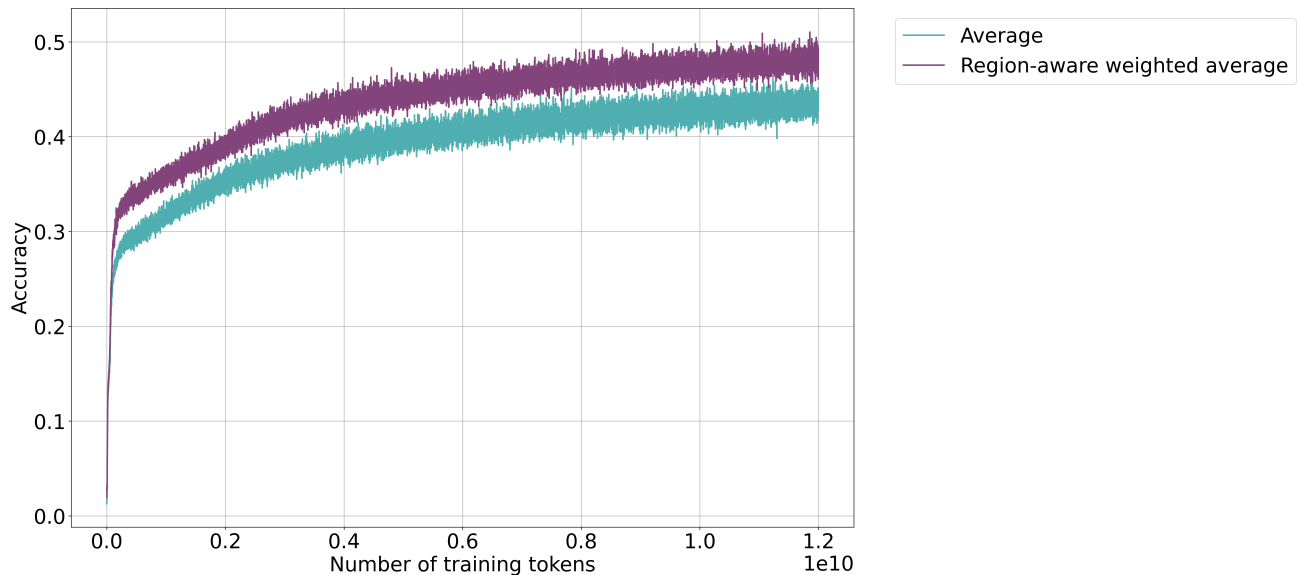


Figure 12. Comparison of *MethFormer* pre-training learning curves for the different methylation preprocessing schemes.

E. Validation on external datasets

We evaluate the generalization of *MOJO* on two cohorts entirely held out from pre-training: the pediatric TARGET dataset and the ICGC cohort. Both datasets were preprocessed using identical pipelines to TCGA and evaluated on the same downstream tasks.

E.1. TARGET

The TARGET (Therapeutically Applicable Research to Generate Effective Treatments, <https://www.cancer.gov/ccg/research/genome-sequencing/target>) dataset comprises primary solid tumor samples from the GDC portal with matched bulk RNA-seq and DNA methylation (Illumina 450k) data and survival information, resulting in 351 patients across four pediatric cancer types.

E.1.1. DATASET COMPOSITION

Table 17. Composition of the TARGET dataset.

Cohort	Cancer type	# Samples	# w/ survival	# Events	# Censored	# Event rate
TARGET-CCSK	Clear Cell Sarcoma of the Kidney	11	11	3	8	27.3%
TARGET-NBL	Neuroblastoma	133	133	66	67	49.6%
TARGET-OS	Osteosarcoma	85	82	29	53	35.4%
TARGET-WT	Wilms Tumor (Kidney)	122	122	50	72	41.0%
Total	All cohorts	351	348	148	200	42.5%

E.1.2. BATCH EFFECT ANALYSIS

We quantified cross-platform batch effects using three complementary metrics: (1) Silhouette scores to measure global dataset separation (range: -1 to 1, with 0 indicating complete mixing), (2) the k -nearest neighbor Batch Effect Test (kBET) to assess whether batch labels are randomly distributed within local neighborhoods (range: 0 to 1, with 1 indicating perfect mixing), and (3) the Pearson correlation of mean bulk RNA-seq and DNA methylation levels between platforms. Our analysis revealed weak batch effects (bulk RNA-seq: Silhouette= 0.19, kBET= 0.95, $r = 0.86$; DNA methylation: Silhouette= 0.12, kBET= 0.92, $r = 0.96$). PCA visualization further showed substantial overlap between TCGA and TARGET samples (Figure 13). The high kBET scores and strong correlations demonstrate excellent local mixing and overall concordance, validating cross-platform generalization.



Figure 13. Bulk RNA-seq and DNA methylation PCA plots — comparison of TCGA and TARGET datasets.

E.1.3. RESULTS

We evaluated models on two tasks: survival analysis (348 patients with follow-up) and zero-shot classification and clustering, discriminating the four cancer types without any fine-tuning on TARGET.

Table 18. External validation on the TARGET dataset.

(a) Survival analysis			(b) Zero-shot classification and clustering		
Model	C-index	Weighted C-index	Metric	MOJO	Late integration
BulkRNABert	0.601 $\pm$ 0.042	0.599 $\pm$ 0.039	Acc.	0.992	0.991
MethFormer	0.548 $\pm$ 0.027	0.540 $\pm$ 0.042	NMI	0.949	0.794
MOJO	0.601 $\pm$ 0.023	0.606 $\pm$ 0.042	ARI	0.946	0.706

E.2. ICGC External Validation Cohort

The ICGC (International Cancer Genome Consortium, <https://dcc.icgc.org>) cohort was assembled from all non-US ICGC projects with publicly available paired bulk RNA-seq and DNA methylation (Illumina 450k) data, after excluding any donors overlapping with TCGA.

E.2.1. DATASET COMPOSITION

Table 19. Composition of the ICGC external validation cohort. Event rate is the proportion deceased among samples with survival information.

Project	Cancer type	# Samples	# w/ survival	# Events	# Censored	Event rate
OV-AU	Ovarian serous cystadenocarcinoma	80	80	66	14	82.5%
PRAD-CA	Prostate adenocarcinoma	142	6	6	0	100.0%
PACA-AU	Pancreatic ductal adenocarcinoma	77	76	46	30	60.5%
CLLE-ES	Chronic lymphocytic leukemia	106	105	27	78	25.7%
Total		405	267	145	122	54.3%

For survival analysis, PRAD-CA was excluded: with only 6 of 142 donors having usable follow-up in ICGC (all 6 deceased), the cohort provides insufficient signal for reliable survival modelling. The remaining three cohorts yield 261 donors with a 53% event rate and a median follow-up of 1,162 days.

E.2.2. RESULTS

We evaluated models on survival analysis (261 donors from OV-AU, PACA-AU, and CLLE-ES), and cancer-type classification (distinguishing the four ICGC lineages across all 405 bimodal samples). Classification performance was uninformative in this setting: all three models trivially reached a macro- F_1 of 1.000, as these four cancer lineages are biologically distant and easily separable by any omics model. The survival results (Table 20) confirm that MOJO’s performance advantage over unimodal baselines generalizes to fully out-of-domain data.

Table 20. Survival analysis on the ICGC external cohort (261 donors; 3 cohorts: OV-AU, PACA-AU, CLLE-ES; 54% event rate). **Bold**: best; underline: second-best.

Model	C-index	Weighted C-index
BulkRNABert	0.805 $\pm$ 0.036	0.610 $\pm$ 0.087
MethFormer	<u>0.810 $\pm$ 0.027</u>	<u>0.628 $\pm$ 0.062</u>
MOJO	0.823 $\pm$ 0.019	0.643 $\pm$ 0.032

F. Interpretability results

In this section, we provide additional results for the interpretability section 5.4. Figure 14 provides a swarm plot of the SHAP values for the *Luminal B* subtype of BRCA. The results demonstrate a strong monotonic impact of the most influential dimensions on *Luminal B* subtype predictions. For a given dimension, its high values are associated with SHAP values of a consistent sign, while its low values are associated with the opposite, indicating that the model has learned distinct predictive patterns. Figure 15 provides the pre-rank GSEA results for all four BRCA subtypes. We only kept the enriched pathways with a false detection rate (FDR) q-value less than 0.25 (Subramanian et al., 2005) and we present only pathways with positive enrichment scores.

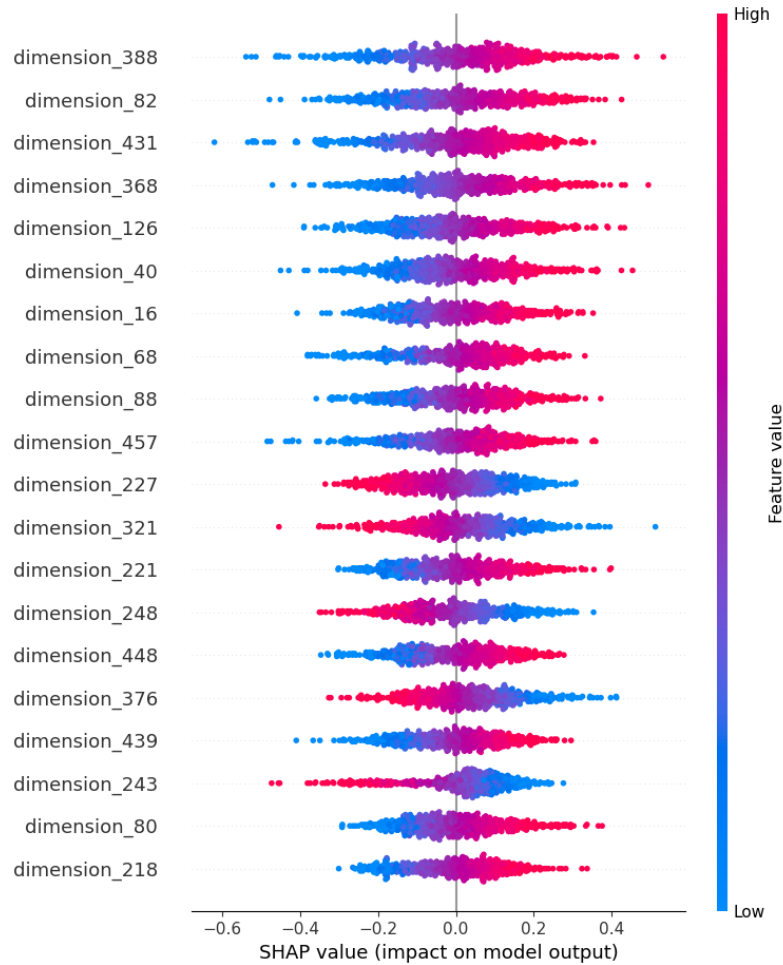


Figure 14. Shapley Values swarmplot for *LumB* sub-type of BRCA cancer.

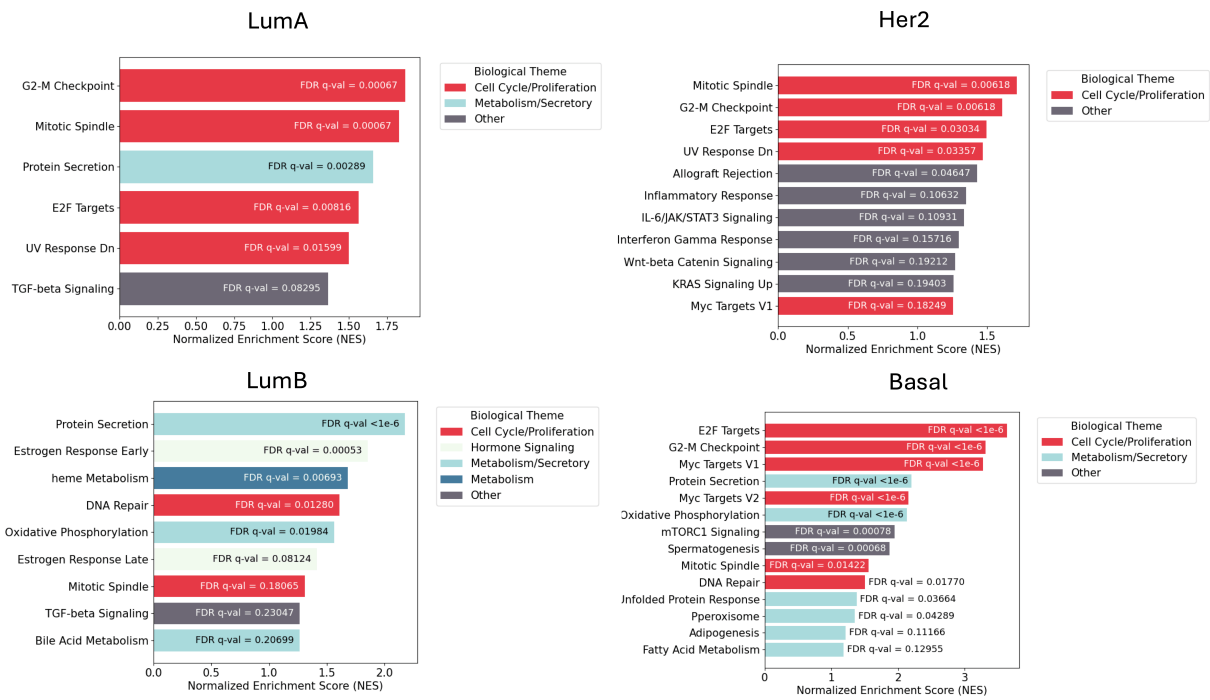


Figure 15. Pre-rank gene set enrichment analysis results for all subtypes of BRCA from bulk RNA-seq correlations with most predictive *MOJO*'s embedding dimensions.

G. Cross-Platform Transferability to 27k Arrays

The Illumina HumanMethylation27k BeadChip is a legacy platform still prevalent in older public cohorts. Unlike the 450k array used to pre-train *MOJO*, 27k arrays cover only $\sim 27,000$ CpG sites with a strong promoter bias, resulting in systematically different gene-level beta-values. Most methylation models trained on 450k data are therefore incompatible with 27k data without retraining. *MOJO*’s gene-level aggregation makes it inherently array-agnostic: switching platforms simply changes which probes contribute to each gene’s mean, without requiring any architectural change. We evaluate this property here and provide the full results supporting Section 6.1.

Dataset construction. We generated an in-silico 27k dataset by filtering our TCGA 450k data to the 27k microarray probe IDs, maintaining exact patients for a 1-to-1 comparison. Gene-level features are computed using the same simple probe averaging as for 450k data. Of the 25,887 27k probes mapping to *MOJO*’s gene vocabulary, 13,726/17,116 genes (80.2%) obtain at least one probe; genes without coverage receive the `<MASK>` token.

Distribution shift quantification. The 27k array’s strong promoter bias causes a systematic mean shift of -0.115 in gene-level beta-values relative to 450k averages, inducing a severe distribution shift that degrades zero-shot model performance.

Evaluation strategies. We compare three adaptation strategies, all with entirely frozen model backbones except for the last:

- *Zero-shot*: the 27k input is fed directly to the model without any adaptation.
- *Rescaled*: for each train/test split, a per-gene z-score normalization is applied using the mean and standard deviation of each covered gene computed over the 450k training fold, re-scaling values to the 450k range. No retraining is required.
- *Head fine-tuning*: task heads are retrained on 27k data while the backbone remains frozen.

Table 21. 27k array transferability under different adaptation strategies for CustOmics and *MOJO*. **Bold**: best per strategy and metric.

Evaluation setting	Model	Cancer-type Classification		Survival Analysis	
		Macro-F1	Weighted-F1	C-index	Weighted C-index
450k (Baseline)	CustOmics	0.922 ± 0.006	0.946 ± 0.006	0.686 ± 0.018	0.639 ± 0.099
	<i>MOJO</i>	0.935 ± 0.007	0.952 ± 0.006	0.771 ± 0.006	0.670 ± 0.009
27k (Zero-shot)	CustOmics	0.226 ± 0.001	0.336 ± 0.002	0.509 ± 0.008	0.515 ± 0.006
	<i>MOJO</i>	0.734 ± 0.024	0.824 ± 0.015	0.717 ± 0.018	0.639 ± 0.025
27k (Rescaled)	CustOmics	0.761 ± 0.012	0.818 ± 0.007	0.633 ± 0.051	0.612 ± 0.044
	<i>MOJO</i>	0.909 ± 0.007	0.940 ± 0.003	0.765 ± 0.007	0.664 ± 0.016
27k (Head fine-tuning)	CustOmics	0.790 ± 0.010	0.837 ± 0.004	0.657 ± 0.054	0.629 ± 0.028
	<i>MOJO</i>	0.913 ± 0.012	0.941 ± 0.005	0.768 ± 0.008	0.660 ± 0.022

The results highlight *MOJO*’s robustness at every level of adaptation. Under zero-shot evaluation, *CustOmics* collapses (macro- F_1 : 0.226) while *MOJO* retains strong performance (0.734), even outperforming fine-tuned *CustOmics* on survival (C-index 0.717 vs. 0.657). Z-score rescaling alone recovers 97% of *MOJO*’s 450k macro- F_1 and 99% of its C-index, with head fine-tuning providing only marginal additional gains. While rescaling may be sensitive to batch effects in real-world deployment, making head fine-tuning the safer practical choice, both strategies confirm that *MOJO* captures highly transferable representations, making legacy 27k cohorts directly usable without re-pre-training or architectural changes.

H. Missing modalities experiments

H.1. Mutual Information computation

Algorithm 1 details the procedure for computing the mutual information loss for a single example. Figure 16 provides the complete results from the missing modalities experiments (when either RNA-seq or methylation is missing), while Table 22 provides exact figures.

Algorithm 1 Mutual information auxiliary (MI) loss

Input: Omics tokens $X = \{rnaseq : x_{rnaseq}, meth : x_{meth}\}$, true class label y , sequence length N , mask token $\langle \text{MASK} \rangle$, mutual information coefficient λ , classification model f_θ

Output: single example loss

if *noMissingModality*(X) **then**

$modalities = [rna + meth, rnaseq, meth]$

$output = [f_\theta(X)]$

for $m \in [rnaseq, meth]$ **do**

$X' \leftarrow copy(X)$

$X'[m] \leftarrow [\langle \text{MASK} \rangle] * N$

$output.append(f_\theta(X'))$

end for

$MILoss = MI(output, modalities)$

else

$MILoss = 0.0$

end if

$Loss = CrossEntropy(f_\theta(X), y) + \lambda * MILoss$

H.2. Experimental results on mutual information

Table 22. Missing modalities experiment: cancer type classification

Model	Add mutual information	Drop modality (test time)	macro-F1	test weighted-F1
BulkRNABert	✗	-	0.918 ± 0.008	0.943 ± 0.004
MethFormer	✗	-	0.917 ± 0.008	0.931 ± 0.006
MOJO	✗	-	0.935 ± 0.007	0.952 ± 0.006
MOJO	✗	Drop 100% of RNA-seq	0.422 ± 0.022	0.538 ± 0.025
MOJO	✗	Drop 100% of Methylation	0.764 ± 0.024	0.854 ± 0.011
MOJO	✓	-	0.930 ± 0.007	0.949 ± 0.004
MOJO	✓	Drop 100% of RNA-seq	0.895 ± 0.008	0.916 ± 0.007
MOJO	✓	Drop 100% of Methylation	0.911 ± 0.012	0.937 ± 0.008
MOJO-Extended-Pretraining	✗	-	0.933 ± 0.006	0.952 ± 0.003
MOJO-Extended-Pretraining	✗	Drop 100% of RNA-seq	0.653 ± 0.013	0.769 ± 0.004
MOJO-Extended-Pretraining	✗	Drop 100% of Methylation	0.903 ± 0.010	0.932 ± 0.005
MOJO-Extended-Pretraining	✓	-	0.929 ± 0.006	0.949 ± 0.005
MOJO-Extended-Pretraining	✓	Drop 100% of RNA-seq	0.883 ± 0.005	0.911 ± 0.004
MOJO-Extended-Pretraining	✓	Drop 100% of Methylation	0.911 ± 0.010	0.937 ± 0.006

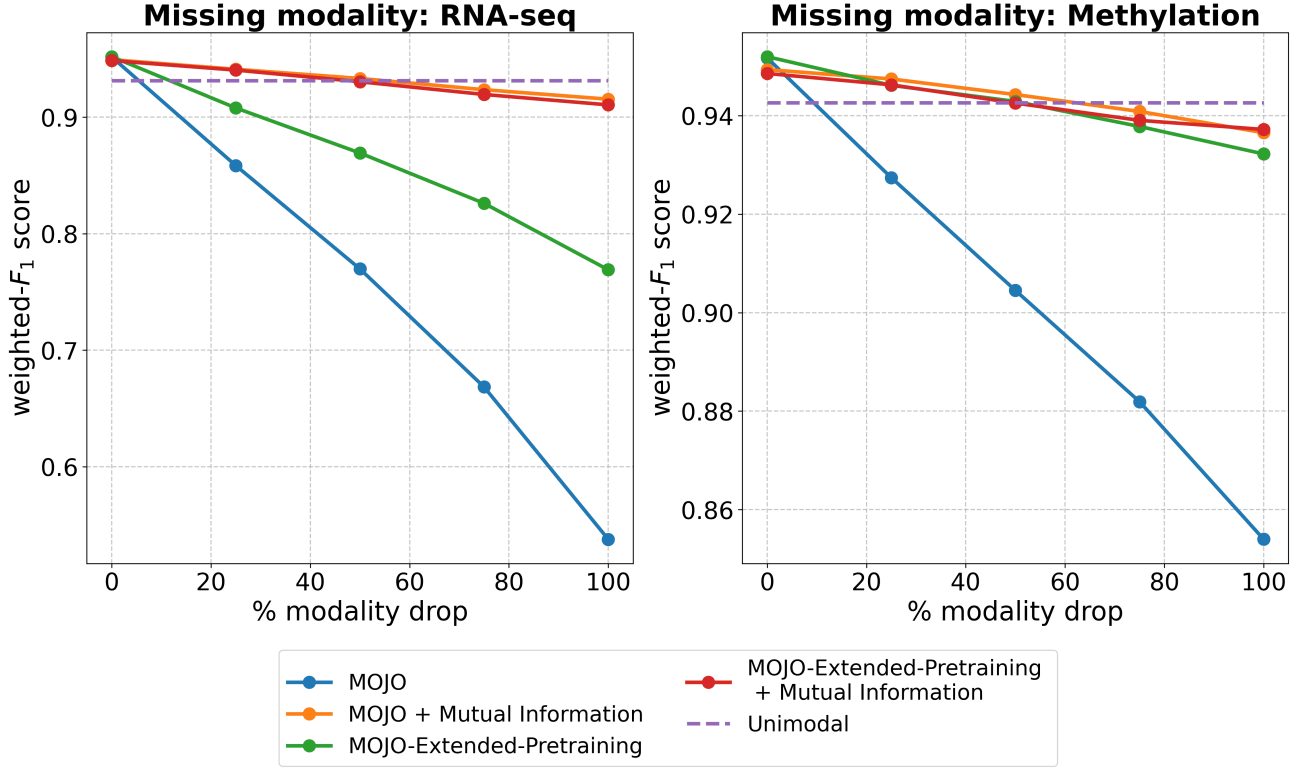


Figure 16. Missing modalities experimental results. Test weighted- F_1 score for the pan-cancer classification is reported for different methods to handle the absence of a modality in $x\%$ of the samples (left: RNA-seq, right: Methylation). Unimodal models are respectively *MethFormer* and *BulkRNABert* when RNA-seq or Methylation is missing.

H.3. Sensitivity analysis: Mutual Information coefficient λ

The missing-modality fine-tuning loss combines a primary cross-entropy term with an auxiliary mutual information term weighted by λ (see Section 7.2). We sweep $\lambda \in \{1, 5, 10, 20, 50\}$ to assess sensitivity to this hyperparameter, evaluating pan-cancer classification weighted- F_1 under full bimodal input and under complete drop of each modality.

Table 23. Sensitivity of the mutual information auxiliary loss to $\lambda \in \{1, 5, 10, 20, 50\}$. Weighted- F_1 on pan-cancer classification under no modality drop, 100% RNA-seq dropped, and 100% methylation dropped.

λ	Weighted-F1 No modality drop	Weighted-F1 Drop 100% RNA-seq	Weighted-F1 Drop 100% methylation
1	0.953 ± 0.006	0.915 ± 0.009	0.939 ± 0.006
5	0.952 ± 0.005	0.918 ± 0.008	0.938 ± 0.007
10	0.949 ± 0.004	0.916 ± 0.007	0.937 ± 0.008
20	0.937 ± 0.006	0.908 ± 0.007	0.924 ± 0.006
50	0.912 ± 0.004	0.889 ± 0.005	0.900 ± 0.006

Performance is stable for $\lambda \leq 10$, confirming our empirical choice. Values $\lambda \geq 20$ degrade performance as the MI loss progressively dominates the primary cross-entropy signal.

H.4. *MOJO* vs. *MOJO-Extended-Pretraining*: full-modality comparison

Figure 4 shows that *MOJO-Extended-Pretraining* improves unimodal fine-tuning (OV subtyping with RNA-seq only). To justify retaining standard *MOJO* as the primary model for bimodal tasks, we benchmark both models when both modalities are available in Table 24:

Table 24. *MOJO* vs. *MOJO-Extended-Pretraining* on full-modality downstream tasks. Results are statistically indistinguishable.

Model	Cancer-type Classification		Survival Analysis	
	Macro-F1	Weighted-F1	C-index	Weighted C-index
<i>MOJO</i>	0.935 ± 0.007	0.952 ± 0.006	0.771 ± 0.006	0.670 ± 0.009
<i>MOJO-Extended-Pretraining</i>	0.933 ± 0.006	0.952 ± 0.003	0.768 ± 0.008	0.666 ± 0.018

Adding unimodal samples during pre-training provides no advantage for bimodal downstream tasks, while it specifically benefits scenarios where one modality is systematically absent during fine-tuning (as in OV sub-typing). We therefore retain standard *MOJO* for all bimodal benchmarks.